

ภาคผนวก ก โครงสร้างไฟล์ Bitmap

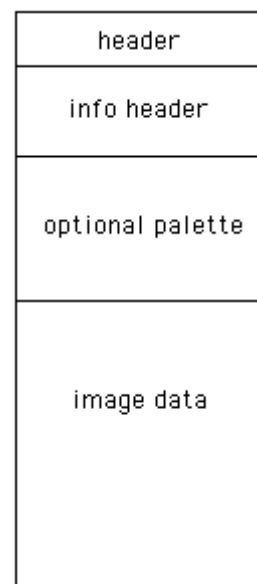
BitMap File

Introduction

BMP files are an historic (but still commonly used) file format for the historic (but still commonly used) operating system called "Windows". BMP images can range from black and white (1 byte per pixel) up to 24 bit colour (16.7 million colours). While the images can be compressed, this is rarely used in practice and won't be discussed in detail here.

Structure

A BMP file consists of either 3 or 4 parts as shown in the diagram on the right. The first part is a header, this is followed by a information section, if the image is indexed colour then the palette follows, and last of all is the pixel data. The position of the image data with respect to the start of the file is contained in the header. Information such as the image width and height, the type of compression, the number of colours is contained in the information header.



1. File Header

The **BITMAPFILEHEADER** structure contains information about the type, size, and layout of a file that contains a DIB.

5 field 14 bytes

Field Name	Size(Byte)	Description
1.bfYype	2	Contains the characters "BM" that identify the file type
2.bfSize	4	Specifies the size, in bytes, of the bitmap file.
3.bfReservaed1	2	Reserved; must be zero
4. bfReservaed2	2	Reserved; must be zero

5.bfOffbits	4	Specifies the offset, in bytes, from the beginning of the BITMAPFILEHEADER structure to the bitmap bits
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2. Image Header

The **BITMAPINFOHEADER** structure contains information about the dimensions and color format of a DIB.

11 field 40 bytes

Field Name	Size(Byte)	Description
1.biSize	4	Specifies the number of bytes required by the structure.
2.biWidth	4	Specifies the width of the bitmap, in pixels.
3. biHeight	4	Specifies the height of the bitmap, in pixels. If biHeight is positive, the bitmap is a bottom-up DIB and its origin is the lower-left corner. If biHeight is negative, the bitmap is a top-down DIB and its origin is the upper-left corner.
4. biplanes	2	Specifies the number of planes for the target device. This value must be set to 1. biBitCount
5. biBitCount	2	Specifies the number of bits-per-pixel. The biBitCount member of the BITMAPINFOHEADER structure determines the number of bits that define each pixel and the maximum number of colors in the bitmap.
6. biCompression	4	Specifies the type of compression for a compressed bottom-up bitmap (top-down DIBs cannot be compressed).
7. biSizeImage	4	Specifies the size, in bytes, of the image. This may be set to zero for BI_RGB bitmaps.
8. biXPelsPerMeter	4	Specifies the horizontal resolution, in pixels-per-meter, of the target device for the bitmap. An application can use this

		value to select a bitmap from a resource group that best matches the characteristics of the current device.
9. biYPelsPerMeter	4	Specifies the vertical resolution, in pixels-per-meter, of the target device for the bitmap.
10. biClrUsed	4	Specifies the number of color indexes in the color table that are actually used by the bitmap. If this value is zero, the bitmap uses the maximum number of colors corresponding to the value of the biBitCount member for the compression mode specified by biCompression .
11. biClrImportant	4	Specifies the number of color indexes that are required for displaying the bitmap. If this value is zero, all colors are required.

biBitCount

Specifies the number of bits-per-pixel. The **biBitCount** member of the **BITMAPINFOHEADER** structure determines the number of bits that define each pixel and the maximum number of colors in the bitmap. This member must be one of the following values.

Value	Meaning
0	Windows 98/Me, Windows 2000/XP: The number of bits-per-pixel is specified or is implied by the JPEG or PNG format.
1	The bitmap is monochrome, and the bmiColors member of BITMAPINFO contains two entries. Each bit in the bitmap array represents a pixel. If the bit is clear, the pixel is displayed with the color of the first entry in the bmiColors table; if the bit is set, the pixel has the color of the second entry in the table.
4	The bitmap has a maximum of 16 colors, and the bmiColors member of BITMAPINFO contains up to 16 entries. Each pixel in the bitmap is represented by a 4-bit index into the color table. For example, if the first byte in the bitmap is 0x1F, the byte represents two pixels. The first pixel contains the color in the second table entry, and the second pixel contains the color in the sixteenth table entry.
8	The bitmap has a maximum of 256 colors, and the bmiColors member of BITMAPINFO contains up to 256 entries. In this case, each byte in the array

represents a single pixel.

- 16 The bitmap has a maximum of 2^{16} colors. If the **biCompression** member of the **BITMAPINFOHEADER** is **BI_RGB**, the **bmiColors** member of **BITMAPINFO** is NULL. Each **WORD** in the bitmap array represents a single pixel. The relative intensities of red, green, and blue are represented with five bits for each color component. The value for blue is in the least significant five bits, followed by five bits each for green and red. The most significant bit is not used. The **bmiColors** color table is used for optimizing colors used on palette-based devices, and must contain the number of entries specified by the **biClrUsed** member of the **BITMAPINFOHEADER**.

If the **biCompression** member of the **BITMAPINFOHEADER** is **BI_BITFIELDS**, the **bmiColors** member contains three **DWORD** color masks that specify the red, green, and blue components, respectively, of each pixel. Each **WORD** in the bitmap array represents a single pixel.

Windows NT/Windows 2000/XP: When the **biCompression** member is **BI_BITFIELDS**, bits set in each **DWORD** mask must be contiguous and should not overlap the bits of another mask. All the bits in the pixel do not have to be used.

Windows 95/98/Me: When the **biCompression** member is **BI_BITFIELDS**, the system supports only the following 16bpp color masks: A 5-5-5 16-bit image, where the blue mask is 0x001F, the green mask is 0x03E0, and the red mask is 0x7C00; and a 5-6-5 16-bit image, where the blue mask is 0x001F, the green mask is 0x07E0, and the red mask is 0xF800.

- 24 The bitmap has a maximum of 2^{24} colors, and the **bmiColors** member of **BITMAPINFO** is NULL. Each 3-byte triplet in the bitmap array represents the relative intensities of blue, green, and red, respectively, for a pixel. The **bmiColors** color table is used for optimizing colors used on palette-based devices, and must contain the number of entries specified by the **biClrUsed** member of the **BITMAPINFOHEADER**.

- 32 The bitmap has a maximum of 2^{32} colors. If the **biCompression** member of the **BITMAPINFOHEADER** is **BI_RGB**, the **bmiColors** member of **BITMAPINFO** is NULL. Each **DWORD** in the bitmap array represents the relative intensities of

blue, green, and red, respectively, for a pixel. The high byte in each **DWORD** is not used. The **bmiColors** color table is used for optimizing colors used on palette-based devices, and must contain the number of entries specified by the **biClrUsed** member of the **BITMAPINFOHEADER**.

If the **biCompression** member of the **BITMAPINFOHEADER** is **BI_BITFIELDS**, the **bmiColors** member contains three **DWORD** color masks that specify the red, green, and blue components, respectively, of each pixel. Each **DWORD** in the bitmap array represents a single pixel.

Windows NT/ 2000: When the **biCompression** member is **BI_BITFIELDS**, bits set in each **DWORD** mask must be contiguous and should not overlap the bits of another mask. All the bits in the pixel do not need to be used.

Windows 95/98/Me: When the **biCompression** member is **BI_BITFIELDS**, the system supports only the following 32-bpp color mask: The blue mask is 0x000000FF, the green mask is 0x0000FF00, and the red mask is 0x00FF0000.

biCompression

Specifies the type of compression for a compressed bottom-up bitmap (top-down DIBs cannot be compressed). This member can be one of the following values.

Value	Description
BI_RGB	An uncompressed format.
BI_RLE8	A run-length encoded (RLE) format for bitmaps with 8 bpp. The compression format is a 2-byte format consisting of a count byte followed by a byte containing a color index. For more information, see Bitmap Compression .
BI_RLE4	An RLE format for bitmaps with 4 bpp. The compression format is a 2-byte format consisting of a count byte followed by two word-length color indexes. For more information, see Bitmap Compression .
BI_BITFIELDS	Specifies that the bitmap is not compressed and that the color table consists of three DWORD color masks that specify the red, green,

and blue components, respectively, of each pixel. This is valid when used with 16- and 32-bpp bitmaps.

BI_JPEG

Windows 98/Me, Windows 2000/XP: Indicates that the image is a JPEG image.

BI_PNG

Windows 98/Me, Windows 2000/XP: Indicates that the image is a PNG image.